

Synergistic Observations of a Galactic Supernova with Super-Kamiokande and Rubin Observatory

The project title:

Synergistic Observations of a Galactic Supernova with Super-Kamiokande and Rubin Observatory

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Brookhaven National Laboratory

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DOE National Laboratory Announcement Number:

LAB 26-3570

DOE/Office of Science Program Office:

High Energy Physics

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COVER PAGE SUPPLEMENT

Provide the following information on a separate page or pages as a Supplement to the Cover Page.

• In the Member List Table, whose format is shown below, list the names, positions, and institutional affiliations of all PIs, Co-Is and senior scientists who will be involved in executing the scopes of work, both U.S. and Japanese, within the international collaboration. The U.S. PI and Japanese PI will be the joint points of contact and coordinators for the combined research activity within each nation. This Table should have its own page(s).

Member List Table

U.S. Collaboration Members			
Number	Name	Affiliation	Position
1	Anže Slosar	Brookhaven National Laboratory	Distinguished Scientist
2	Chris Walter	Duke University	Professor
3	Dan Scolnic	Duke University	Associate Professor
4	Peter Denton	BNL	Research Staff
5	Robert Blum	NOIRLab	Director of Rubin Observations
6	Philip Marshall	SLAC	Deputy Director of Rubin Observations
7	Sandrine Thomas	NOIRLAB	Associate Director for Rubin Summit Operations
8	Aaron Roodman	SLAC	Deputy Head of LSST

Japanese Collaboration Members			
Number	Name	Institution	Position
1	Hiroyuki Sekiya	Kamioka Obs., UTokyo	Associate Professor
2	Guillaume Pronost	Kamioka Obs., UTokyo	Project Assistant Professor
3	Motoyasu Ikeda	Kamioka Obs., UTokyo	Assistant Professor
4	Yousuke Utsumi	NAOJ	Associate Professor
5	Masaomi Tanaka	Tohoku University	Professor
6	Nozomu Tominaga	NAOJ	Professor
7	Masayuki Nakahata	Kamioka Obs., UTokyo	Senior Fellow

- The Budget Requests Table below collects into a single table all individual institutions' budgets. Continuing the example given in Section I, "Multi-Institutional Teams," assume the total project cost listed in DOE Lab A's budget is \$300,000, of which \$50,000, \$40,000 and \$90,000 are to be provided as subawards by DOE Lab A to DOE Lab B, and Universities C and D, respectively. The Budget Requests Table should then list the amounts \$120,000, \$50,000, \$40,000 and \$90,000 for DOE Lab A, DOE Lab B, University C and University D, respectively. The budget total would be listed as \$300,000.
- In addition to these numbers, please provide the collaboration's total multi-institutional request to KEK, broken down into Equipment and Supplies Costs and Travel Costs. These tabulated data are planned to be used by DOE and KEK for coordination purposes.

Example budget table (in U.S.\$)

Budget Request to DOE (in USD)				
Name	Institution	FY2026 Request	FY2027 Request	FY2028 Request
Anze Slosar	Brookhaven National Laboratory	100,000	100,000	
TOTAL		100,000	100,000	
Budget Request to KEK (in kJPY)				
Equipment and Supplies Costs		¥	¥	¥
Travel Costs		¥2500	¥2500	¥

Abstract

When a supernova (SN) occurs in our Galaxy, neutrino observatories will detect a neutrino burst hours before the explosion is visible in optical bands. Super-Kamiokande can play a unique and central role in this effort: it is currently the only detector capable of efficiently determining the neutrino-based pointing direction for a Galactic supernova using neutrino–electron elastic scattering. This directional information is essential for enabling prompt and effective optical follow-up observations. At the same time, the high sensitivity and large field of view of the Vera C. Rubin Observatory makes it an ideal instrument to follow such neutrino triggers for Target-of-Opportunity (ToO) observations, providing rapid optical follow-up and initial localization.

In the work to date we have demonstrated the feasibility of Rubin-led follow-up observations triggered by neutrino detectors, engaged the Rubin ToO community, and produced a concrete set of recommendations for implementation [1]. The current performance of the Super-Kamiokande early alert system, including its directional sensitivity, is summarized in [2].

The purpose of the present project is to advance this work from concept to implementation by incorporating these recommendations into the Rubin Observatory ToO infrastructure. This will include developing the necessary interface code, implementing software support for real-time image differencing and supernova detection, and carrying out an end-to-end validation campaign. The resulting infrastructure will also provide a foundation for future integration of additional neutrino detectors, such as DUNE. Super-Kamiokande will provide real-time neutrino triggers, directional information, and operational expertise in low-latency alert generation. Close coordination with the Japanese Rubin community will ensure that the neutrino and optical systems are coherently integrated.

To ensure the success of this proposal, one of the U.S. PIs is funded by the DOE Office of Science (HEP) and is an active member of both the Rubin and Super-Kamiokande collaborations. On the Japanese side, the project involves key members of the Super-Kamiokande leadership—including the Super-Kamiokande spokesperson and senior detector and analysis experts at the Kamioka Observatory—as well as leading members of the Japanese Rubin community from NAOJ and Japanese universities. Their involvement ensures close coordination between neutrino observations, alert generation, and optical follow-up strategies.

The primary deliverable of this project is a fully integrated, validated, and tested infrastructure that enables the Rubin Observatory to ingest and act upon real-time neutrino triggers from Super-Kamiokande (and other future detectors). A secondary but significant outcome is the strengthening of the scientific community and the development of a robust science case around neutrino-triggered optical observations of Galactic supernovae. This will be supported through an international workshop in Japan near the Super-Kamiokande site, as well as focused meetings in the United States.

Project Narrative

Executive Summary

This exploratory application is a tightly focused two-year proposal to successfully implement a Rubin Observatory target of opportunity observing system to see the light from a galactic supernova after a neutrino-based warning from Super-Kamiokande. The pointing capabilities of Super-K are almost exactly matched to the field of view of Rubin whose depth will also allow us to peer deeply into the dust of the galactic center. This work will be coupled with a unique international workshop in Japan to develop the best follow-up strategies in all optical observatories. This proposed work for the DOE-supported Super-Kamiokande and Rubin Observatories will act as R&D towards follow-up for the next generation neutrino detectors and systems coming online during the decade, including DUNE, Hyper-Kamiokande, SNEWS-II, and Juno.

Introduction

Approximately once per century, a core-collapse supernova explodes in a typical galaxy like our own [3, 4]. Core-collapse supernovae happen when a massive star (heavier than about 8–10 solar masses) starts to run out of the hydrogen that has been fueling its fusion burning [5]. The resulting loss of pressure leads to a contraction of the star, raising its temperature until the burning of the helium produced in the previous hydrogen fusion can begin. This cycle repeats, next burning carbon, neon, oxygen, and silicon until finally a core of iron remains [6]. The loss of pressure from fusion causes a sudden collapse; this collapse is halted when nuclear densities are reached and, at this point, the supernova shock forms and the neutrinos begin to stream towards Earth [7]. (The detection of the neutrino burst from SN 1987A provided direct confirmation that core collapse produces an intense, seconds-long neutrino signal [8, 9].) After this, the ultimate fate of the star is determined by a battle between the accreting matter and the intense outward pressures from heat, neutrinos, and turbulence: in the delayed neutrino-heating picture of [10], multi-dimensional instabilities and neutrino heating can help revive the stalled shock [7]. If the former wins, a black hole is formed; if the latter wins, the supernova shock drives out into the star, starting an explosion [11, 12]. Depending on the size of the star, minutes to hours later, the shock wave will actually break out of the stellar envelope and become visible as a supernova explosion to optical telescopes [13, 14].

Studying a galactic supernova in detail would be an **incredible opportunity to do fundamental physics** [15, 16]: it probes the behavior of macroscopic objects at nuclear density [17, 18], it probes the physics of gravity at densities that are high enough to (maybe) trigger black hole formation [11, 12], it probes neutrino physics at densities that we can never replicate on Earth—including flavor conversion effects en route to the detector [15, 19, 20]—and it probes fundamental interactions since any new channel for energy loss would appear as a deficit and/or shortening of the neutrino signal [16, 21]. In short, catching a supernova in action would be transformational for physics in general and would directly touch essentially all facets of high-energy and nuclear physics [7, 15].

We expect such events to happen in our vicinity approximately once or twice per century [22]. There are a handful of events observed and recorded by humans over the past few millennia in the form of a new temporary star appearing in the night sky. The only confirmed core collapse observed in modern history was SN 1987A in the nearby LMC. The optical aftershock of this supernova has been detected by Las Campanas Observatory in Chile [23, 24] and Kamiokande-II, IMB, and Baksan have (post-facto) confirmed an excess of about 24 neutrinos a few hours before the optical detection has taken place [25–27]. This demonstrated that the basic physics is well understood. The 1987A

supernova has been scientifically enormously impactful: NASA ADS identifies over 5000 papers written about 1987A and data has been taken with various telescopes decades after the explosion has been first detected.

In the past three decades there has been a massive improvement in our ability to measure neutrinos and analyze the data in real time. Therefore, if another supernova goes off in our vicinity, we expect neutrino detectors to be the first to detect it. Super-Kamiokande will detect this initial neutrino shower and localize the supernova to an area of about a few degrees across in the sky. This is where the Vera Rubin Observatory comes into play. The large field of view and the large collecting area of the Simonyi telescope are perfectly positioned to survey the sky around the neutrino-determined region for a few hours after the neutrino trigger to catch the first optical afterglow following the neutrinos. There are other telescopes with larger mirrors, but none matches the unique combination of a multi-square-degree field of view and observation availability provided by the DOE-developed LSST camera and a sufficient sensitivity to detect the optical afterglow even when supernova’s light is heavily extinguished by the intervening dust. After the new supernova has been located and confirmed by Rubin, its location will be known to better than arcsecond precision, setting off a subsequent trigger that will train an entire arsenal of more powerful photometric and spectroscopic facilities at all wavelengths to study the object in greater detail. At the same time, Rubin will measure the very important early light-curve from the emerging supernova. After the initial localization and early light-curve characterization, Rubin, with its large field of view, is poorly suited for the detailed follow up and will return to performing routine survey operations.

Progress to Date

Scenario described above has been common lore in the community for a while. It has been advocated in a white paper submitted to an open call for the LSST cadence optimization committee [28], on which several proposers were co-authors. In FY24 and FY25, this work has been supported by the Lab Directed Research & Development (LDRD) grant at Brookhaven National Laboratory (BNL) which supported a large fraction of a postdoc.

During this period, the Rubin project has started actively moving towards a better understanding of various Target-of-Opportunity (ToO) observations that LSST is well suited to do. This class of observations is dominated by the gravitational-wave community, which can provide a predictable and relatively large number of sources for a potential follow-up by the Rubin Observatory. A dedicated ToO workshop has been organized, which resulted in a community whitepaper “Rubin ToO 2024: Envisioning the Vera C. Rubin Observatory LSST Target of Opportunity program” [34], in which a galactic supernova is presented in a dedicated section on this science case. This section has been co-edited by the postdocs supported by the BNL program.

In parallel, we have started to address the basic practical questions associated with this science opportunity, focusing on the Rubin follow-up side. There are two main sets of questions

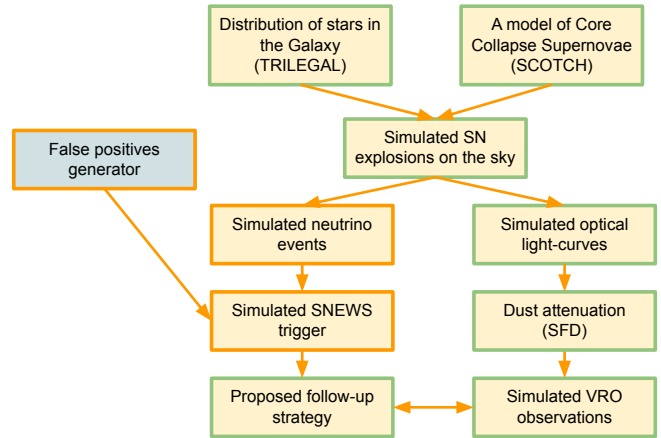


Figure 1: This figure schematically shows the ingredients of a simulation suite. Boxes labelled in green were implemented in [1]. We use the TRILEGAL [29–31] catalog to simulate the distribution of stars, the SCOTCH [32] library of SN light curves, and the SFD map for the distribution of Galactic dust [33].

associated with this problem. The first set of questions deals with the fundamental issues of observability of core-collapse supernovae by Rubin. What fraction of objects do we lose because they are too far north to be observed by Rubin? What fraction of objects do we lose because they are too reddened by dust in the Galactic center? Conversely, how likely is a supernova so bright that it might trigger a detector trip-off? To answer these and related questions like this, we need a realistic simulation scheme, shown schematically in Figure 1. In [1] we have implemented this on the Rubin side and made predictions for the distribution of observed supernova magnitudes as well as concrete recommendations for the observing strategy given the neutrino trigger. The main conclusions of this work were encouraging: a majority of SN occurring in our own Galaxy should be observable by LSST and, in the reddest filter, essentially all of them are detectable in the standard Rubin 30 s exposure, despite very significant reddening, particularly in the Galactic center.

The second set of questions concerns the ability to detect a new source close to the Galactic plane. While Rubin has been designed as a time-domain-focused instrument from the very beginning, sources in the very dense stellar fields close to the Galactic center pose particular problems. Crowded stellar fields imply numerous variable stars and therefore numerous false positives, the source differencing algorithms have not been tested in dense stellar fields and might require special fine-tuning, and there are uncertainties surrounding the availability of templates near the Galactic center, etc. In order to answer these questions, we have studied the performance of the Rubin differencing algorithms in very crowded stellar fields, representative of those near the Galactic plane [35]. We have identified the major obstacles for an efficient detection of a supernova explosion: i) events are lost to bleed trails which are omnipresent in the dense stellar fields and ii) the several-magnitude decrease in the signal-to-noise, which is dominated by unresolved stars rather than sky background. With short exposures and focal plane rotation the first issue can be mitigated while still providing sufficient sensitivity to detect all supernova in red bands.

The work to date has established a baseline on which to implement the actual software stack necessary to implement the neutrino trigger. The coarse architecture of the system that enables a Rubin ToO observation is well understood and has been tested in realistic conditions for gravitational-wave events. It is shown in Figure 2. The alert coming from Super-Kamiokande first needs to be transformed into a SCIMMA alert, which is the format that is ingested by the piece of software running on the mountain called the Rubin ToO producer. This producer injects the alert, potentially merges it with additional information and automatically decides on whether to proceed with observations. For events which are produced with some regularity, such as GW alerts, the ToO producer might decide to skip them if conditions are not favorable. A credible galactic supernova alert certainly warrants observations as long as it is feasible (i.e. during the night on an accessible patch of the sky, not too close to the Moon, etc.). The ToO producer defines the observation area (based on Super-Kamiokande’s initial localization) and observing parameters (such as integration time, filters, etc.), which are passed to the so-called Feature-Based Scheduler (FBS) which, given the high ToO priority, interrupts the regular survey operations and executes the ToO observations. FBS is responsible for tiling of the area and actual running of the telescope. At this point, we envision a human operator entering the loop. The hu-

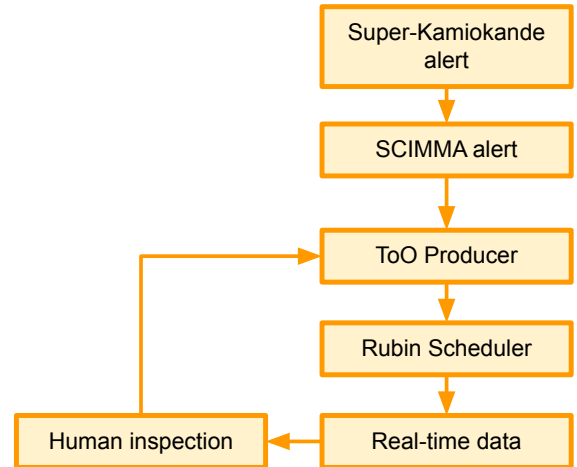


Figure 2: The flow of information from the original Super-Kamiokande trigger to the Rubin observations.

Timeline (months)	Goal	Description
1-6	Ensure timely production and processing of SCIMMA alerts.	We will ensure that the Super-Kamiokande and SNEWS alerts are propagated through the SCIMMA alert system and picked up by Rubin with appropriately short latency.
1-6	Implement Galactic SN processing into ToO producer.	While a placeholder implementation exists, we will work with the Rubin team to fill in the blanks and implement a fully functional code that covers all corner cases.
6-12	Implement the real-time analysis scripts.	We will implement scripts that will be used by human operators during the ToO observations to assess candidate events and deal with corner cases such as detector trip-off.
6-18	Simulate Super-Kamiokande events.	The simulation pipeline has been largely implemented from the Rubin side (see Figure 1). We will implement simulation from the Super-Kamiokande side to better understand the localization error as a function of supernova position and trigger time;
12-24	Perform end-to-end testing and validation of the whole pipeline.	The infrastructure for performing end-to-end testing is available within Rubin during engineering (daytime) operations, whereby real raw images with a synthetic source injection can be fed into the system in lieu of actual observations.
1-24	Organize a set of workshops in Japan and US to organize the community.	We plan to organize two meetings: one in Toyama, Japan, near the Super-Kamiokande site and another in Tucson, USA, home of NOIR-Lab, which runs the Rubin Observatory. These workshops will focus on bringing together both the neutrino and astronomy communities to better understand the implications of a Galactic supernova event and novel ways of using such an event to further our understanding of fundamental physics. We plan to have two sets of internal team-only meetings (one US and one Japan) and two workshops with wider participation while still maintaining a working meeting environment (15–30 participants).

Table 1: Goals and timeline of the proposed work.

man operator will look at the output of the ToO pipeline and manually confirm candidate events, after which the event data are edited to focus observations at the position of the real source.

Proposed Work & Timetable

The main purpose of this proposal is to develop all the necessary middleware to process a Galactic supernova neutrino alert, validate and test it in realistic conditions, and obtain a better understanding of the system capability, its limitations, and various failure modes. At the same time, we want to bring together both communities that have been thinking about Galactic core-collapse supernovae under one roof to understand the implications of such an event and the subsequent analysis. While Super-Kamiokande is the most sensitive instrument of the current generation, more detectors such as DUNE [36] and Hyper-K [37] are expected to come online over the next decade. This makes the groundwork proposed here even more important.

In Table 1 we enumerate the goals and the timeline for this project.

Team & Resources

We have assembled an experienced team that contains internationally recognized experts in this field. On the US side, our team involves all the co-PIs on the BNL LDRD involved in exploratory

work: Anže Slosar, the BNL PI who also serves on the Rubin Science Advisory Committee and DESC Spokesperson Advisory Committee; Chris Walter (Duke), who was intimately involved in Rubin commissioning and is also a member of Super-Kamiokande; Dan Scolnic (Duke), who is a world leader in time-domain astronomy with deep expertise in image subtraction; and Peter Denton (BNL), a neutrino theorist who straddles cosmology, HEP, and nuclear physics. To this team we are adding technical Rubin experts: Robert Blum (NOIRLab), who is Rubin Director of Operations, his deputy Phil Marshal, (SLAC), Sandrine Thomas, who is Associate Director for Rubin Summit Operations and Aaron Roodman (SLAC), who led the LSST Camera Development at SLAC and is now Rubin Deputy Head of LSST. The direct involvement of LSST leadership will enable efficient resource allocation required for testing and trigger response optimization.

On the Japanese side, our team combines strong expertise in neutrino detection, real-time supernova alerts, and time-domain astronomy. Hiroyuki Sekiya (Kamioka Observatory, Tokyo) is a serving Super-Kamiokande Collaboration Spokesperson and has played a leading role in the SK-Gd project and the development of supernova neutrino alert systems. Guillaume Pronost and Motoyasu Ikeda (Kamioka Observatory, Tokyo) are conveners of the Super-Kamiokande supernova working group and are experts in the construction and operation of real-time alert pipelines based on SK data. Yousuke Utsumi (NAOJ) is an astronomer with in-depth knowledge of the Rubin Observatory through his extensive involvement in Rubin commissioning. The team also includes Masaomi Tanaka (Tohoku) and Nozomu Tominaga (NAOJ), both experts in theoretical and observational studies of supernovae and transients, with extensive experience in electromagnetic follow-ups of multi-messenger signals; Tominaga additionally serves as a liaison to the Japanese small-telescope network OISTER, which is crucial for bright Galactic supernovae. Masayuki Nakahata (Kamioka Observatory, Tokyo), former Director of the Institute for Cosmic Ray Research and former Super-Kamiokande Spokesperson, adds long-standing leadership and experience in large-scale neutrino experiments.

We ask for resources to cover 0.4 FTE of a postdoc involved in performing the software tasks required for this work. We additionally request resources that will enable working visits between the US and Japan as well as the organization of workshops. We hope to be able to cover workshop travel for junior members of the community. Our workshop in Toyama will involve a visit to Super-Kamiokande, while the workshop in Tucson will showcase NOIR facilities, where Rubin has been developed.

Conclusions

We propose a novel project that builds upon existing work. If successful, the project will pave the way for an optimal and efficient follow-up by the Rubin observatory of a galactic core-collapse supernovae. This project is a high-risk - high-reward project in the most fundamental sense of the work – the probability of a core-collapse supernova occurring during the next ten years is approximately 10%. However, even if supernova does not go off in our galaxy in the next decade, this work will pave the way for subsequent observations by more sensitive instruments coming online in the future – Rubin observatory will continue to operate past its LSST Survey and Super-Kamiokande will be followed up by Hyper-Kamiokande and DUNE. Galactic supernovae are one of the DUNE's primary science objectives.

An observation of a core-collapse supernovae in our galaxy would address a number of questions in fundamental physics and is well aligned with the DOE HEP and KEK objectives. Funding this project will build new ties between Japanese and US HEP communities and new ties between neutrino physicists from the US intensity frontiers and cosmologist from the US cosmic frontier. We have an optimal team of experts that are ready to turn this into reality.

Appendix 1: Biographical Sketches

IDENTIFYING INFORMATION:

NAME: Slosar, Anže

PERSISTENT IDENTIFIER (PID): <https://orcid.org/0000-0002-8713-3695>

POSITION TITLE: Distinguished Scientist

PRIMARY ORGANIZATION AND LOCATION: Brookhaven National Laboratory, Upton, New York, United States

Professional Preparation:

ORGANIZATION AND LOCATION	DEGREE (if applicable)	RECEIPT DATE	FIELD OF STUDY
University of Cambridge, Cambridge, Not Applicable, N/A, United Kingdom	PHD	10/2000 - 07/2003	Physics
University of Cambridge, Cambridge, Not Applicable, N/A, United Kingdom	MS	10/1996 - 06/2000	Theoretical and Experimental Physics

Appointments and Positions

2009 - present Distinguished Scientist, Brookhaven National Laboratory, Upton, New York, United States

2007 - 2009 Inaugural Fellow, Berkeley Center for Cosmological Physics, Berkeley, California, United States

2005 - 2007 Research Assistant, University of Oxford, Oxford, Not Applicable, N/A, United Kingdom

2003 - 2005 University of Oxford, University of Ljubljana, Ljubljana, Not Applicable, N/A, Slovenia

Products

1. Dvorkin C., Dvorkin C., Gerbino M., Alonso D., Battaglia N., Bird S., Rivero A.D., Font-Ribera A., Fuller G., Lattanzi M., Loverde M., Mu\~noz J.B., Sherwin B., Slosar A., Villaescusa-Navarro F., Abazajian K.N., Abidi M., Adshead P., Amin M.A., Ansarinejad B., Armstrong R., Asorey J., Avila S., Baccigalupi C., Barron D., Bechtol K., De Belsunce R., Bennett C., Benson B., Bernal J.L., Beutler F., Bilicki M., Biviano A., Blazek J., Richard Bond J., Borrill J., Buckley-Geer E., Bull P., Burgess C., Calabrese E., Castorina E., Chaves-Montero J., Comparat J., Croft R.A.C., Cyr-Racine F.-Y., D'Amico G., Davis T.M., Delabrouille J., Dor\~e O., Drlica-Wagner A., Ellison J., Essinger-Hileman T., Ferraro S., Ferreira P.G., Flauger R., Foreman S., Fosalba P., Bouchet F.R., Garc\'ia-Bellido J., Garc\'ia-Bellido J., Gill M.S.S., Gluscevic V., Gontcho S.G.A., Green D., Grohs E., Gruen D., Gupta N., Hahn C., Hanany S., Hawken A.J., Colin Hill J., Hirata C.M., Hlo\~ek R., Holder G., Horiuchi S., Huterer D., Ishak M., Jeltema T., Kamionkowski M., Keeley R.E., Knox L., Koushiappas S.M., Kovetz E.D., Koyama K., L'Huillier B., Lahav O., Leonard D., Liguori M., Liu A., Liu J., De La Macorra A., Mancini A.S., Manera M., Mantz A., Martini P., Massara E., McQuinn M., Daniel Meerburg P., Meyers J., Miralda-Escud\~e J., Miranda V., Mirbabayi M., More S., Myers A.D., Palanque-Delabrouille N., Newburgh L., Niemack M.D., Niz G., Percival W.J., Piacentini F., Piacentini

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Certification:

I understand that I have been designated as a covered individual by the Federal funding agency.

I certify to the best of my knowledge and belief that the information contained in this Biographical Sketch Form is true, complete, and accurate. I understand that any false, fictitious, or fraudulent information, misrepresentations, half-truths, or omissions of any material fact, may subject me to criminal, civil or administrative penalties for fraud, false statements, false claims or otherwise. (18 U.S.C. §§ 1001 and 287, and 31 U.S.C. 3729-3733 and 3801-3812). I further understand and agree that (1) the statements and representations made herein are material to DOE's funding decision, and (2) I have a responsibility to update the disclosures during the project period of the award should circumstances change which impact the responses provided above.

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Certified by Slosar, Anže in SciENCv on 2026-01-20 05:02:41

IDENTIFYING INFORMATION:

NAME: Walter, Christopher

PERSISTENT IDENTIFIER (PID): <https://orcid.org/0000-0003-2035-2380>

POSITION TITLE: Chair of Department of Physics

PRIMARY ORGANIZATION AND LOCATION: Duke University, Durham, North Carolina, United States

Professional Preparation:

ORGANIZATION AND LOCATION	DEGREE (if applicable)	RECEIPT DATE	FIELD OF STUDY
California Institute of Technology, Pasadena, CA, USA	PHD	06/1997	Physics
California Institute of Technology, Pasadena, CA, USA	MS	06/1991	Physics
University of California at Santa Cruz, Santa Cruz, CA, USA	BA	06/1989	Physics

Appointments and Positions

2025 - present Chair of Department of Physics, Duke University, Durham, North Carolina, United States

2017 - present Professor of Physics, Duke University, Durham, NC, United States

2008 - present Visiting Scientist, Institute for Physics and Mathematics of the Universe, University of Tokyo, Kashiwa, Not Applicable, N/A, Japan

2011 - 2017 Associate Professor of Physics, Duke University, Durham, NC, USA

2004 - 2010 Assistant Professor of Physics, Duke University, Durham, NC, USA

2000 - 2003 Research Assistant Professor, Boston University, Boston, MA, USA

1997 - 2000 Postdoctoral Research Associate, Boston University, Boston, MA, USA

Products

1. Amon A., Bechtol K., Connolly A.J., Digel S.W., Drlica-Wagner A., Gawiser E., Jarvis M., Jha S.W., von der Linden A., Moniez M., Narayan G., Regnault N., Sevilla-Noarbe I., Schmidt S.J., Suyu S.H., Walter C.W.. Recommended Target Fields for Commissioning the Vera C. Rubin Observatory. 2020 October. other-id: 1826870
2. Abolfathi B, et al. The LSST DESC DC2 Simulated Sky Survey. The Astrophysical Journal Supplement Series. 2021 March 12; 253(1):31-. Available from: <https://iopscience.iop.org/article/10.3847/1538-4365/abd62c> DOI: 10.3847/1538-4365/abd62c
3. Walter Christopher W., Scolnic Daniel M., Slosar Anže. LSST Target of Opportunity proposal for locating a core collapse supernova in our galaxy triggered by a neutrino supernova alert. . other-id: 1712754
4. Guo Z, Walter C, Lage C, Lupton R. Fringing Analysis and Simulation for the Vera C. Rubin Observatory's Legacy Survey of Space and Time. Publications of the Astronomical Society of

the Pacific. 2023 March 27; 135(1045):034503-. Available from:
<https://iopscience.iop.org/article/10.1088/1538-3873/acbe67> DOI: 10.1088/1538-3873/acbe67

5. Longley E, Chang C, Walter C, Zuntz J, Ishak M, Mandelbaum R, Miyatake H, Nicola A, Pedersen E, Pereira M, Prat J, Sánchez J, Secco L, Tröster T, Troxel M, Wright A. A unified catalogue-level reanalysis of stage-III cosmic shear surveys. Monthly Notices of the Royal Astronomical Society. 2023 April; 520(4):5016-5041. Available from:
<https://academic.oup.com/mnras/article/520/4/5016/7008908> DOI: 10.1093/mnras/stad246

Certification:

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I certify to the best of my knowledge and belief that the information contained in this Biographical Sketch Form is true, complete, and accurate. I understand that any false, fictitious, or fraudulent information, misrepresentations, half-truths, or omissions of any material fact, may subject me to criminal, civil or administrative penalties for fraud, false statements, false claims or otherwise. (18 U.S.C. §§ 1001 and 287, and 31 U.S.C. 3729-3733 and 3801-3812). I further understand and agree that (1) the statements and representations made herein are material to DOE's funding decision, and (2) I have a responsibility to update the disclosures during the project period of the award should circumstances change which impact the responses provided above.

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Certified by Walter, Christopher in SciENcv on 2026-01-14 17:37:39

IDENTIFYING INFORMATION:

NAME: Scolnic, Daniel

ORCID iD: <https://orcid.org/0000-0002-4934-5849>

POSITION TITLE: Associate Professor

PRIMARY ORGANIZATION AND LOCATION: Duke University, Durham, North Carolina, United States**Professional Preparation:**

ORGANIZATION AND LOCATION	DEGREE (if applicable)	RECEIPT DATE	FIELD OF STUDY
Johns Hopkins University, Baltimore, Maryland, United States	PHD	09/2007 - 09/2013	Physics
Johns Hopkins University, Baltimore, Maryland, United States	MS	09/2003 - 09/2007	Physics
Massachusetts Institute of Technology, Cambridge, Massachusetts, United States	BS	09/2003 - 05/2007	Physics

Appointments and Positions

2023 - present Associate Professor, Duke University, Durham, North Carolina, United States

2025 - 2027 Associate Professor, Duke University, Durham, North Carolina, United States

2019 - 2023 Assistant Professor, Duke University, Durham, North Carolina, United States

2015 - 2019 Hubble/KICP Postdoctoral Fellow, The University of Chicago, Chicago, Illinois, United States

2013 - 2015 Research Scientist, Johns Hopkins University, Baltimore, Maryland, United States

2007 - 2013 Graduate Researcher, Johns Hopkins University, Baltimore, Maryland, United States

Products**Products Most Closely Related to the Proposed Project**

1. Sanchez B. The Dark Energy Survey Supernova Program: light curves and 5-year data release. The Astrophysical Journal. 2024 November 01; 975.
2. Vincenzi M. The Dark Energy Survey Supernova Program: cosmological analysis and systematic uncertainties. The Astrophysical Journal. 2024 October 25; 975.
3. DES Collaboration . The Dark Energy Survey: cosmology results with ~1500 new high-redshift Type Ia supernovae using the full 5-year dataset. The astrophysical journal. Letters. 2024 October 01; 973.
4. Abbott T. First cosmology results using Type Ia supernovae from the Dark Energy Survey: constraints on cosmological parameters. The astrophysical journal. Letters. 2021 February 21; 872.

Other Significant Products, Whether or Not Related to the Proposed Project

1. Brout D. First Cosmology Results Using SNe Ia from the Dark Energy Survey: Analysis, Systematic Uncertainties, and Validation. The Astrophysical Journal. 2019 April 02; 874.

Certification:

I certify that the information provided is current, accurate, and complete. This includes, but is not limited to, information related to current, pending, and other support (both foreign and domestic) as defined in 42 U.S.C. § 6605.

In accordance with Section 10632 of the CHIPS and Science Act of 2022 (42 U.S.C. § 19232), each individual identified as a senior/key person must certify that they are not a party to a malign foreign talent recruitment program.

Research Security Training Requirement for Federal Award Personnel: In accordance with Section 10634 of the CHIPS and Science Act of 2022 (42 U.S.C. § 19234), each individual identified as a senior/key person must certify that they have completed the requisite research security training that meets the requirements specified in Item 2 of Important Notice No. 149 within 12 months prior to proposal submission.

Certified by Scolnic, Daniel in SciENCv on 2026-01-17 15:37:41

IDENTIFYING INFORMATION:

NAME: Denton, Peter

PERSISTENT IDENTIFIER (PID): <https://orcid.org/0000-0002-5209-872X>

POSITION TITLE: Physicist

PRIMARY ORGANIZATION AND LOCATION: Brookhaven National Lab, Upton, New York, United States

Professional Preparation:

ORGANIZATION AND LOCATION	DEGREE (if applicable)	RECEIPT DATE	FIELD OF STUDY
Vanderbilt University, Nashville, Tennessee, United States	PHD	08/2010 - 08/2016	Physics
Rice University, Houston, Texas, United States	BA	08/2006 - 05/2010	Math
Rice University, Houston, Texas, United States	BS	08/2006 - 05/2010	Physics

Appointments and Positions

2018 - present Physicist, Brookhaven National Lab, Upton, New York, United States

2016 - 2018 Postdoc, Niels Bohr Institute, Copenhagen, Not Applicable, N/A, Denmark

Products

1. Denton PB. Probing CP Violation with Neutrino Disappearance Alone. Phys Rev Lett. 2024 Jul 19;133(3):031801. PubMed PMID: [39094130](#).
2. Denton PB. Sterile Neutrino Search with MicroBooNE's Electron Neutrino Disappearance Data. Phys Rev Lett. 2022 Aug 5;129(6):061801. PubMed PMID: [36018666](#).
3. Davoudiasl H, Denton PB, Gehrlein J. Supermassive Black Holes, Ultralight Dark Matter, and Gravitational Waves from a First Order Phase Transition. Phys Rev Lett. 2022 Feb 25;128(8):081101. PubMed PMID: [35275682](#).
4. Denton PB, Gehrlein J, Pestes R. CP-Violating Neutrino Nonstandard Interactions in Long-Baseline-Accelerator Data. Phys Rev Lett. 2021 Feb 5;126(5):051801. PubMed PMID: [33605742](#).
5. Cui Y, Yu J, de Roeck A, Sousa A, de Gouvea A, Denton P, Machado PAN. New Opportunities at the Next-Generation Neutrino Experiments (Part 1: BSM Neutrino Physics and Dark Matter. Rep Prog Phys. 2020 Jun 15; PubMed PMID: [32541096](#).
6. Davoudiasl H, Denton PB. Ultralight Boson Dark Matter and Event Horizon Telescope Observations of M87^{*}. Phys Rev Lett. 2019 Jul 12;123(2):021102. PubMed PMID: [31386502](#).
7. Denton PB, Tamborra I. Invisible Neutrino Decay Could Resolve IceCube's Track and Cascade Tension. Phys Rev Lett. 2018 Sep 21;121(12):121802. PubMed PMID: [30296122](#).

Certification:

I understand that I have been designated as a covered individual by the Federal funding agency.

I certify to the best of my knowledge and belief that the information contained in this Biographical Sketch Form is true, complete, and accurate. I understand that any false, fictitious, or fraudulent information, misrepresentations, half-truths, or omissions of any material fact, may subject me to criminal, civil or administrative penalties for fraud, false statements, false claims or otherwise. (18 U.S.C. §§ 1001 and 287, and 31 U.S.C. 3729-3733 and 3801-3812). I further understand and agree that (1) the statements and representations made herein are material to DOE's funding decision, and (2) I have a responsibility to update the disclosures during the project period of the award should circumstances change which impact the responses provided above.

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Certified by Denton, Peter in SciENCv on 2026-01-14 09:01:13

IDENTIFYING INFORMATION:

NAME: Blum, Robert David

PERSISTENT IDENTIFIER (PID): <https://orcid.org/0000-0002-8622-4237>

POSITION TITLE: Director

PRIMARY ORGANIZATION AND LOCATION: NSF-DOE Vera C. Rubin Observatory/NSF NOIRLab, Tucson, Arizona, United States

Professional Preparation:

ORGANIZATION AND LOCATION	DEGREE (if applicable)	RECEIPT DATE	FIELD OF STUDY
The Ohio State University, Columbus, Ohio, United States	PHD	09/1990 - 05/1995	Astronomy
University of Washington, Seattle, Washington, United States	MS	09/1985 - 06/1986	Aeronautics and Astronautics
University of Washington, Seattle, Washington, United States	BS	09/1981 - 08/1985	Aeronautics and Astronautics

Appointments and Positions

2025 - present	Director , NSF-DOE Vera C. Rubin Observatory/NSF NOIRLab, Tucson, Arizona, United States
2011 - present	Full Astronomer, NOAO/NOIRLab, Tucson, Arizona, United States
2021 - 2025	Director for Operations, NOIRLab/Rubin Observatory, Tucson, Arizona, United States
2019 - 2021	Director, Vera C. Rubin Observatory Operations (Acting), Rubin Observatory/NSF's NOIRLab, Tucson, Arizona, United States
2009 - 2018	Deputy Director, NOAO, Tucson, Arizona, United States
2008 - 2009	Interim Deputy Director, NOAO, Tucson, Arizona, United States
2006 - 2012	Manager, Telescope System Instrumentation Program, NOAO, Tucson, Arizona, United States
2006 - 2008	Associate Astronomer, NOAO Gemini Science Center, NOAO, Tucson, Arizona, United States
2002 - 2006	Associate Astronomer, CTIO (tenure, May 2004), CTIO, La Serena, Not Applicable, N/A, Chile
1997 - 2002	Assistant Astronomer, CTIO, CTIO, La Serena, Not Applicable, N/A, Chile
1995 - 1997	Hubble Postdoctoral Fellow, University of Colorado, JILA, Boulder, Colorado, United States
1993 - 1994	Presidential Fellowship, The Ohio State University, Columbus , Ohio, United States
1991 - 1993	Perkins Research Assistantship, The Ohio State University, Columbus , Ohio, United States

Products

1. Dey A, Schlegel D, Lang D, Blum R, et al. Overview of the DESI Legacy Imaging Surveys. *The Astronomical Journal*. 2019 April 11; 157(5):168-. Available from: <https://iopscience.iop.org/article/10.3847/1538-3881/ab089d> DOI: 10.3847/1538-3881/ab089d
2. Meixner M, Gordon K, Indebetouw R, Hora J, Whitney B, Blum R, Reach W, Bernard J, Meade M, Babler B, Engelbracht C, For B, Misselt K, Vijh U, Leitherer C, Cohen M, Churchwell E, Boulanger F, Frogel J, Fukui Y, Gallagher J, Gorjian V, Harris J, Kelly D, Kawamura A, Kim S, Latter W, Madden S, Markwick-Kemper C, Mizuno A, Mizuno N, Mould J, Nota A, Oey M, Olsen K, Onishi T, Paladini R, Panagia N, Perez-Gonzalez P, Shibai H, Sato S, Smith L, Staveley-Smith L, Tielens A, Ueta T, van Dyk S, Volk K, Werner M, Zaritsky D. Spitzer Survey of the Large Magellanic Cloud: Surveying the Agents of a Galaxy's Evolution (SAGE). I. Overview and Initial Results. *The Astronomical Journal*. 2006 December 1; 132:2268-2288. Available from: <https://ui.adsabs.harvard.edu/abs/2006AJ....132.2268M> DOI: 10.1086/508185
3. Nidever D, Olsen K, Walker A, Vivas A, Blum R, Kaleida C, Choi Y, Conn B, Gruendl R, Bell E, Besla G, Muñoz R, Gallart C, Martin N, Olszewski E, Saha A, Monachesi A, Monelli M, de Boer T, Johnson L, Zaritsky D, Stringfellow G, van der Marel R, Cioni M, Jin S, Majewski S, Martinez-Delgado D, Monteagudo L, Noël N, Bernard E, Kunder A, Chu Y, Bell C, Santana F, Frechem J, Medina G, Parkash V, Navarrete J, Hayes C. SMASH: Survey of the MAGellanic Stellar History. *The Astronomical Journal*. 2017 November 1; 154:199. Available from: <https://ui.adsabs.harvard.edu/abs/2017AJ....154..199N> DOI: 10.3847/1538-3881/aa8d1c
4. Blum R, Ramírez S, Sellgren K, Olsen K. Really Cool Stars and the Star Formation History at the Galactic Center. *The Astrophysical Journal*. 2003 November 1; 597:323-346. Available from: <https://ui.adsabs.harvard.edu/abs/2003ApJ...597..323B> DOI: 10.1086/378380
5. DESI Collaboration, Aghamousa A, Aguilar J, ..., Blum R, et al.. The DESI Experiment Part I: Science, Targeting, and Survey Design. 2016 October. Available from: <https://ui.adsabs.harvard.edu/abs/2016arXiv161100036D> DOI: 10.48550/arXiv.1611.00036
6. Moon J, Valcin D, Rashkovetskyi M, Saulder C, Aguilar J, Ahlen S, Alam S, Bailey S, Baltay C, Blum R, Brooks D, Burtin E, Chaussidon E, Dawson K, de la Macorra A, de Mattia A, Dhungana G, Eisenstein D, Flaugher B, Font-Ribera A, Forero-Romero J, Garcia-Quintero C, Gontcho A Gontcho S, Guy J, Hanif M, Honscheid K, Ishak M, Kehoe R, Kim S, Kisner T, Kremin A, Landriau M, Le Guillou L, Levi M, Manera M, Martini P, McDonald P, Meisner A, Miquel R, Moustakas J, Myers A, Nadathur S, Neveux R, Newman J, Nie J, Padmanabhan N, Palanque-Delabrouille N, Percival W, Pérez Fernández A, Poppett C, Prada F, Raichoor A, Ross A, Rossi G, Samushia L, Schlegel D, Seo H, Tarlé G, Vargas Magana M, Variu A, Weaver B, White M, Yèche C, Yuan S, Zhao C, Zhou R, Zhou Z, Zou H. First detection of the BAO signal from early DESI data. *Monthly Notices of the Royal Astronomical Society*. 2023 November 1; 525:5406-5422. Available from: <https://ui.adsabs.harvard.edu/abs/2023MNRAS.525.5406M> DOI: 10.1093/mnras/stad2618
7. Bianco F, Ivezić Ž, Jones R, Graham M, Marshall P, Saha A, Strauss M, Yoachim P, Ribeiro T, Anguita T, Bauer A, Bauer F, Bellm E, Blum R, Brandt W, Brough S, Catelan M, Clarkson W, Connolly A, Gawiser E, Gizis J, Hložek R, Kaviraj S, Liu C, Lochner M, Mahabal A, Mandelbaum R, McGehee P, Neilsen Jr., Olsen K, Peiris H, Rhodes J, Richards G, Ridgway S, Schwamb M, Scolnic D, Shemmer O, Slater C, Slosar A, Smartt S, Strader J, Street R, Trilling

D, Verma A, Vivas A, Wechsler R, Willman B. Optimization of the Observing Cadence for the Rubin Observatory Legacy Survey of Space and Time: A Pioneering Process of Community-focused Experimental Design. The Astrophysical Journal Supplement Series. 2022 January 1; 258:1. Available from: <https://ui.adsabs.harvard.edu/abs/2022ApJS..258....1B> DOI: 10.3847/1538-4365/ac3e72

8. Adame A. G., Aguilar J., Ahlen S.. DESI 2024 III: baryon acoustic oscillations from galaxies and quasars. Journal of Cosmology and Astroparticle Physics. 2025 April. bibcode: 2025JCAP...04..012A
9. Adame A. G., Aguilar J., Ahlen S.. DESI 2024 VI: cosmological constraints from the measurements of baryon acoustic oscillations. Journal of Cosmology and Astroparticle Physics. 2025 February. bibcode: 2025JCAP...02..021A
10. Adame A. G., Aguilar J., Ahlen S.. DESI 2024 VII: cosmological constraints from the full-shape modeling of clustering measurements. Journal of Cosmology and Astroparticle Physics. 2025 July. bibcode: 2025JCAP...07..028A
11. Palaversa L., Donev E., Ivezić Ž.. photoD with Rubin's Data Preview 1: first stellar photometric distances and deficit of faint blue stars. Stellar distances with Rubin's DP1. 2025 December. bibcode: 2025arXiv251224109P
12. Malanchev Konstantin, DeLucchi Melissa, Caplar Neven. Variability-finding in Rubin Data Preview 1 with LSDB. 2025 June. bibcode: 2025arXiv250623955M

Certification:

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Certified by Blum, Robert David in SciENCv on 2026-01-17 10:40:13

IDENTIFYING INFORMATION:

NAME: Thomas, Sandrine J

ORCID iD: <https://orcid.org/0000-0002-9121-3436>

POSITION TITLE: Associate Director for Rubin Summit Operation

PRIMARY ORGANIZATION AND LOCATION: AURA / NOIRLab / Vera C. Rubin Observatory, Tucson, Arizona, United States

Professional Preparation:

ORGANIZATION AND LOCATION	DEGREE (if applicable)	RECEIPT DATE	FIELD OF STUDY
Université Nice Sophia Antipolis, Nice, Not Applicable, N/A, FR	PhD in Science of the Universe	09/2002 - 12/2005	Astrophysics
Institut d'Optique Graduate School, Orsay, Not Applicable, N/A, FR	Optical Engineer	09/1998 - 06/2001	Optics

Appointments and Positions

2025 - present Associate Director for Rubin Summit Operation, AURA / NOIRLab / Vera C. Rubin Observatory, Tucson, Arizona, United States

2015 - 2025 Deputy Director / Telescope Scientist, AURA / Vera C. Rubin Observatory, Tucson, Arizona, United States

2012 - 2015 Research Associate, NASA Ames, Mountain View, California, United States

2011 - 2012 Gemini Planet Imager (GPI) Instrument Specialist , Gemini, Santa Cruz, California, United States

2008 - 2011 Research Scientist, UCSC/Laboratory for Adaptive Optics, Santa Cruz, California, United States

2006 - 2008 Post-doctoral Researcher, UCSC/Laboratory for Adaptive Optics, Santa Cruz, California, United States

Products**Products Most Closely Related to the Proposed Project**

1. Crenshaw J, Connolly A, Meyers J, Kalmbach J, Megias Homar G, Ribeiro T, Suberlak K, Thomas S, Tsai T. Using AI for Wave-front Estimation with the Rubin Observatory Active Optics System. The Astronomical Journal. 2024 January 31; 167(2):86-. Available from: <https://iopscience.iop.org/article/10.3847/1538-3881/ad1661> DOI: 10.3847/1538-3881/ad1661
2. Homar Guillem Megias, Kahn Steven M, Meyers Joshua M, Crenshaw John Franklin, Thomas Sandrine J. The Active Optics System on the Vera C. Rubin Observatory: Optimal Control of Degeneracy Among the Large Number of Degrees of Freedom. arXiv preprint arXiv:2406.04656. 2024.
3. Thomas Sandrine, Connolly Andrew, Crenshaw John Franklin, Bryce Kalmbach J, Megias Guillem, Meyers Joshua E, Ribeiro Tiago, Tsai Te-Wei, Claver Chuck, Neill Doug, others. Rubin Observatory Simonyi Survey Telescope Active Optics. Adaptive Optics for Extremely

Large Telescopes 7th Edition; 2023; c2023.

4. Xin Bo, Neill Doug R, Thomas Sandrine, Claver Chuck F, Contaxis Christopher, Hileman Ed, Hubler William W, Loeff Adrian R, Kenagy Kurtis L, Martin Hubert M, others. Rubin Observatory primary tertiary mirror cell assembly: Optical testing of the fully assembled mirror support. Ground-based and Airborne Telescopes VIII; 2020; c2020.
5. Thomas Sandrine J, Barr Jeffrey, Callahan Shawn, Clements Andy W, Daruich Felipe, Fábrega Juan, Ingraham Patrick, Muñoz Freddy, Neill Douglas, Ribeiro Tiago, others. Rubin Observatory Simonyi survey telescope status overview. Ground-based and Airborne Telescopes IX; 2022; c2022.

Other Significant Products, Whether or Not Related to the Proposed Project

1. Thomas Sandrine J, Gill Ranpal, Shugart Alysha, Connolly Andrew, Dubois Richard, Daruich Felipe, Chirino Carol, Corlies Lauren, Blum Robert, Bianco Federica, others. Creating an inclusive and diverse environment at Vera C. Rubin Observatory. Observatory Operations: Strategies, Processes, and Systems IX; 2022; c2022.
2. Ingraham Patrick, Fagrelus Parker, Stubbs Christopher W, Lupton Robert H, Liang Ming, Neill Doug, Muller Gary, Claver Charles F, Sebag Jacques, Thomas Sandrine, others. The Vera C. Rubin Observatory 8.4 m telescope calibration system status. Ground-based and Airborne Telescopes IX; 2022; c2022.
3. Thomas Sandrine, Tranco Gelys, Vernet Elise, Travouillon Tony. Special Section Guest Editorial: Extremely Large Telescopes. Journal of Astronomical Telescopes, Instruments, and Systems. 2022; 8(2):021501--021501.
4. Thomas S, Rodgers B, van der Blik N, Doppmann G, Bouvier J, Salvo C, Beuzit J, Rigaut F. A Survey of Herbig Ae/Be Multiplicity. The Astronomical Journal. 2023 February 28; 165(3):135-. Available from: <https://iopscience.iop.org/article/10.3847/1538-3881/aca803> DOI: 10.3847/1538-3881/aca803
5. Sebag Jacques, Claver Charles F, Thomas Sandrine J, Reil Kevin, Barr Jeffrey, Bechtol Keith, Daruich Felipe, Fabrega Juan, Guy Leanne P, Ingraham Patrick, others. Vera C. Rubin Observatory system integration, test, and commissioning: strategy and status. Ground-based and Airborne Telescopes VIII; 2020; c2020.

Certification:

I certify that the information provided is current, accurate, and complete. This includes, but is not limited to, information related to current, pending, and other support (both foreign and domestic) as defined in 42 U.S.C. § 6605.

In accordance with Section 10632 of the CHIPS and Science Act of 2022 (42 U.S.C. § 19232), each individual identified as a senior/key person must certify that they are not a party to a malign foreign talent recruitment program.

Research Security Training Requirement for Federal Award Personnel: In accordance with Section 10634 of the CHIPS and Science Act of 2022 (42 U.S.C. § 19234), each individual identified as a senior/key person must certify that they have completed the requisite research security training that meets the requirements specified in Item 2 of Important Notice No. 149 within 12 months prior to

proposal submission.

Certified by Thomas, Sandrine J in SciENcv on 2026-01-20 07:00:14

IDENTIFYING INFORMATION:

NAME: Roodman, Aaron

PERSISTENT IDENTIFIER (PID): <https://orcid.org/0000-0001-5326-3486>

POSITION TITLE: Professor

PRIMARY ORGANIZATION AND LOCATION: SLAC National Accelerator Laboratory, Stanford University, Department of Particle Physics & Astrophysics, Menlo Park, California, United States**Professional Preparation:**

ORGANIZATION AND LOCATION	DEGREE (if applicable)	RECEIPT DATE	FIELD OF STUDY
Enrico Fermi Institute, University of Chicago, Chicago, Illinois, United States	Postdoctoral Fellow	08/1991 - 06/1994	Particle Physics
University of Chicago, Chicago, Illinois, United States	PHD	09/1985 - 09/1991	Physics
California Institute of Technology, Pasadena, California, United States	BS	09/1981 - 06/1985	Physics

Appointments and Positions

2012 - present	Professor, SLAC National Accelerator Laboratory, Stanford University, Department of Particle Physics & Astrophysics, Menlo Park, California, United States
2005 - 2012	Associate Professor, SLAC National Accelerator Laboratory, Stanford University, Department of Particle Physics & Astrophysics, Menlo Park, California, United States
1998 - 2005	Assistant Professor, Stanford Linear Accelerator Center, Stanford University, Department of Particle Physics & Astrophysics, Menlo Park, California, United States
1994 - 1998	Research Scientist, Enrico Fermi Institute, University of Chicago, Chicago, Illinois, United States

Products

1. Park J, Birrer S, Ueland M, Cranmer M, Agnello A, Wagner-Carena S, Marshall P, Roodman A. Hierarchical Inference of the Lensing Convergence from Photometric Catalogs with Bayesian Graph Neural Networks. The Astrophysical Journal. 2023 August 16; 953(2):178-. Available from: <https://iopscience.iop.org/article/10.3847/1538-4357/acdc25> DOI: 10.3847/1538-4357/acdc25
2. Park J, Wagner-Carena S, Birrer S, Marshall P, Lin J, Roodman A. Large-scale Gravitational Lens Modeling with Bayesian Neural Networks for Accurate and Precise Inference of the Hubble Constant. The Astrophysical Journal. 2021 March 24; 910(1):39-. Available from: <https://iopscience.iop.org/article/10.3847/1538-4357/abdfc4> DOI: 10.3847/1538-4357/abdfc4
3. Wagner-Carena S, Park J, Birrer S, Marshall P, Roodman A, Wechsler R. Hierarchical Inference with Bayesian Neural Networks: An Application to Strong Gravitational Lensing. The

Astrophysical Journal. 2021 March 17; 909(2):187-. Available from:
<https://iopscience.iop.org/article/10.3847/1538-4357/abdf59> DOI: 10.3847/1538-4357/abdf59

4. Ivezić Ž, Kahn S, Tyson J, et.al., LSST: From Science Drivers to Reference Design and Anticipated Data Products. The Astrophysical Journal. 2019 March 11; 873(2):111-. Available from: <https://iopscience.iop.org/article/10.3847/1538-4357/ab042c> DOI: 10.3847/1538-4357/ab042c
5. Roodman A, Reil K, Davis C. Wavefront sensing and the active optics system of the dark energy camera. In: Stepp L, Gilmozzi R, Hall H, editors. SPIE Astronomical Telescopes + Instrumentation; Sund 2 ; Montréal, Quebec, Canada. Available from: <http://proceedings.spiedigitallibrary.org/proceeding.aspx?doi=10.1117/12.2056904> DOI: 10.1117/12.2056904

Certification:

I understand that I have been designated as a covered individual by the Federal funding agency.

I certify to the best of my knowledge and belief that the information contained in this Biographical Sketch Form is true, complete, and accurate. I understand that any false, fictitious, or fraudulent information, misrepresentations, half-truths, or omissions of any material fact, may subject me to criminal, civil or administrative penalties for fraud, false statements, false claims or otherwise. (18 U.S.C. §§ 1001 and 287, and 31 U.S.C. 3729-3733 and 3801-3812). I further understand and agree that (1) the statements and representations made herein are material to DOE's funding decision, and (2) I have a responsibility to update the disclosures during the project period of the award should circumstances change which impact the responses provided above.

I also certify that, at the time of submission, I am not a party in a malign foreign talent recruitment program. I further understand should I take action to involve myself with a Malign Foreign Talent Recruitment Program during the period of performance of the award, I must notify the recipient's Authorized Agent immediately, but no later than five business days of taking such action and immediately recuse myself from all DOE awards.

Certified by Roodman, Aaron in SciENCv on 2026-01-26 11:44:00

Appendix 3: Current and Pending Support

CURRENT AND PENDING (OTHER) SUPPORT INFORMATION

Provide the following information for the Senior/key personnel and other significant contributors.
Follow this format for each person.

*NAME: Slosar, Anže

PERSISTENT IDENTIFIER (PID): <https://orcid.org/0000-0002-8713-3695>

*POSITION TITLE: Distinguished Scientist

*ORGANIZATION AND LOCATION: Brookhaven National Laboratory, Upton, New York, United States

Proposals and Active Projects

*Proposal/Active Project Title: Brookhaven National Laboratory

*Status of Support: Current

Proposal/Award Number: B&R KA2301020/FWP PO205 (Ongoing)

*Source of Support: DOE Office of Science High Energy Physics

*Primary Place of Performance: Brookhaven National Laboratory

*Proposal/Active Project Start Date: (MM/YYYY): 10/2023

*Proposal/Active Project End Date: (MM/YYYY): 09/2028

*Total Anticipated Proposal/Project Amount: \$500,000

* Person Months per budget period Devoted to the Proposal/Active Project:

Year	Person Months
2026	5.4
2027	6
2028	6

*Overall Objectives: LuSEE-Night Science

*Statement of Potential Overlap: No overlap.

*Proposal/Active Project Title: Cosmic Frontier Research

*Status of Support: Current

Proposal/Award Number: &R KA2301020/FWP PO104

***Source of Support:** DOE Office of Science High Energy Physics

***Primary Place of Performance:** Brookhaven National Laboratory

***Proposal/Active Project Start Date: (MM/YYYY):** 10/2023

***Proposal/Active Project End Date: (MM/YYYY):** 09/2028

***Total Anticipated Proposal/Project Amount:** \$1,060,000

*** Person Months per budget period Devoted to the Proposal/Active Project:**

Year	Person Months
2026	4.8
2027	4.2
2028	4.2

***Overall Objectives:** Base grant.

***Statement of Potential Overlap:** Limited overlap.

***Proposal/Active Project Title:** Cosmic Frontier Research

***Status of Support:** Current

Proposal/Award Number: B&R KA2401043/FWP PO104 (Ongoing)

***Source of Support:** DOE Office of Science High Energy Physics

***Primary Place of Performance:** Brookhaven National Laboratory

***Proposal/Active Project Start Date: (MM/YYYY):** 10/2023

***Proposal/Active Project End Date: (MM/YYYY):** 09/2028

***Total Anticipated Proposal/Project Amount:** \$130,000

*** Person Months per budget period Devoted to the Proposal/Active Project:**

Year	Person Months
2026	1.8
2027	1.8
2027	1.8

***Overall Objectives:** AI/ML part of the base grant.

***Statement of Potential Overlap:** No overlap.

Addendum

Do the Notice of Funding Opportunity (NOFO) or Award instructions require a Current and Pending (Other) Support addendum?

Yes

1. Have you started or spun-out any companies, whether foreign, domestic, publicly traded, or non-publicly traded?

No

2. Do you have an affiliation with an institution located in an FCOC?

No

3. Within the past seven years, have you collaborated with individuals at an FCOC site or used facilities or instrumentation at an FCOC site? Collaborations do not include routine workshops or conferences hosted in FCOCs.

No

4. Do you currently serve, or have you ever served, in the military or intelligence service or police force of an FCOC?

No

5. Within the past seven years, have you been offered any of the following by an FCOC: a. Future compensation; b. Complimentary foreign travel, lodging, and/or per diem; c. Things of non de minimus value (i.e., when aggregated, exceeds \$5,000); d. Honorific titles; e. Career advancement opportunities; f. Other types of remuneration or consideration

No

6. Within the past seven years have you received any FCOC sources of support, whether from a public or private entity or individual? Sources of support include:

- a. Salary and any payment for services not otherwise identified as salary (e.g., consulting fees, honoraria, paid authorship);**
- b. Equity interest (e.g., any stock, stock option, or other ownership interest);**
- c. Income related to intellectual property rights and interests (e.g., patents, copyrights);**
- d. Reimbursed or sponsored travel (i.e., that which is paid on behalf of the covered individual and not reimbursed to the covered individual so that the exact monetary value may not be readily available);**
- e. Current and pending (other) support as defined in the National Security Presidential Memorandum-33 Implementation Guidance Appendix Definitions;**
- f. Non-monetary support of goods, services, or expertise;**
- g. Payments from an FCOC to family or friends on the covered individual's behalf or held in any other accounts on their behalf (promise of future access).**

No

7. To the best of your knowledge, are you an inventor on a patent application filed in an

FCOC, without a companion US patent application?

No

Certification:

I understand that I have been designated as a covered individual by the Federal funding agency.

I certify to the best of my knowledge and belief that the information contained in this Current and Pending (Other) Support Form is true, complete, and accurate. I understand that any false, fictitious, or fraudulent information, misrepresentations, half-truths, or omissions of any material fact, may subject me to criminal, civil or administrative penalties for fraud, false statements, false claims or otherwise. (18 U.S.C. §§ 1001 and 287, and 31 U.S.C. 3729-3733 and 3801-3812). I further understand and agree that (1) the statements and representations made herein are material to DOE's funding decision, and (2) I have a responsibility to update the disclosures during the project period of the award should circumstances change which impact the responses provided above.

I also certify that, at the time of submission, I am not a party in a malign foreign talent recruitment program. I further understand should I take action to involve myself with a Malign Foreign Talent Recruitment Program during the period of performance of the award, I must notify the recipient's Authorized Agent immediately, but no later than five business days of taking such action and immediately recuse myself from all DOE awards.

If this is an R&D proposal/award, I also certify to the following statement:

Within the past 12 months I have completed research security training meeting the requirements in SEC. 10634(b) of 42 USC 19234.

Certified by Slosar, Anže in SciENCv on 2026-01-20 05:13:07

CURRENT AND PENDING (OTHER) SUPPORT INFORMATION

Provide the following information for the Senior/key personnel and other significant contributors.
Follow this format for each person.

*NAME: Walter, Christopher

PERSISTENT IDENTIFIER (PID): <https://orcid.org/0000-0003-2035-2380>

*POSITION TITLE: Professor

*ORGANIZATION AND LOCATION: Duke University, Durham, North Carolina, United States

Proposals and Active Projects

*Proposal/Active Project Title: Research in High Energy Physics at Duke University

*Status of Support: Current

Proposal/Award Number: DE-SC0010007

*Source of Support: DOE

*Primary Place of Performance: Duke University

*Proposal/Active Project Start Date: (MM/YYYY): 04/2025

*Proposal/Active Project End Date: (MM/YYYY): 03/2028

*Total Anticipated Proposal/Project Amount: \$5,996,000

* Person Months per budget period Devoted to the Proposal/Active Project:

Year	Person Months
2025	2
2026	2
2027	2

*Overall Objectives: Research in High Energy Physics at Duke University. I am overall PI of our umbrella grant which includes 12 faculty, and also CO-PI on task C: Cosmology.

*Statement of Potential Overlap: The work in the US-Japan proposal is related to this work, but I am not asking for any extra funds.

Addendum

Do the Notice of Funding Opportunity (NOFO) or Award instructions require a Current and Pending (Other) Support addendum?

No

Certification:

I understand that I have been designated as a covered individual by the Federal funding agency.

I certify to the best of my knowledge and belief that the information contained in this Current and Pending (Other) Support Form is true, complete, and accurate. I understand that any false, fictitious, or fraudulent information, misrepresentations, half-truths, or omissions of any material fact, may subject me to criminal, civil or administrative penalties for fraud, false statements, false claims or otherwise. (18 U.S.C. §§ 1001 and 287, and 31 U.S.C. 3729-3733 and 3801-3812). I further understand and agree that (1) the statements and representations made herein are material to DOE's funding decision, and (2) I have a responsibility to update the disclosures during the project period of the award should circumstances change which impact the responses provided above.

I also certify that, at the time of submission, I am not a party in a malign foreign talent recruitment program. I further understand should I take action to involve myself with a Malign Foreign Talent Recruitment Program during the period of performance of the award, I must notify the recipient's Authorized Agent immediately, but no later than five business days of taking such action and immediately recuse myself from all DOE awards.

If this is an R&D proposal/award, I also certify to the following statement:

Within the past 12 months I have completed research security training meeting the requirements in SEC. 10634(b) of 42 USC 19234.

Certified by Walter, Christopher in SciENCv on 2026-01-16 13:40:14

CURRENT AND PENDING (OTHER) SUPPORT INFORMATION

Provide the following information for the Senior/key personnel and other significant contributors.
Follow this format for each person.

*NAME: Scolnic, Daniel

*POSITION TITLE: Professor of Physics

*ORGANIZATION AND LOCATION: Duke University, Durham, North Carolina, United States

Proposals/Active Projects

*Proposal/Active Project Title: Roman Space Telescope High Latitude Supernova Program

*Status of Support: Current

Proposal/Award Number:

*Source of Support: NASA

*Primary Place of Performance: Duke University, Durham, North Carolina, United States

*Proposal/Active Project Start Date: (MM/YYYY): 10/2023

*Proposal/Active Project End Date: (MM/YYYY): 10/2028

*Total Anticipated Proposal/Project Amount: \$11,000,000

* Person Months per budget period Devoted to the Proposal/Active Project:

Year	Person Months
2024	1
2025	1
2026	1
2027	1
2028	1

*Overall Objectives: Build Infrastructure for the upcoming Roman Space Mission

*Statement of Potential Overlap: This work is highly Roman-specific, though some of the software may be utilized by DESC

***Proposal/Active Project Title:** Research in High Energy Physics at Duke University (Task C)

***Status of Support:** Current

Proposal/Award Number:

***Source of Support:** DOE

***Primary Place of Performance:** Duke University, Durham, North Carolina, United States

***Proposal/Active Project Start Date: (MM/YYYY):** 04/2022

***Proposal/Active Project End Date: (MM/YYYY):** 03/2025

***Total Anticipated Proposal/Project Amount:** \$1,526,791

*** Person Months per budget period Devoted to the Proposal/Active Project:**

Year	Person Months
2022	1
2023	1
2024	1

***Overall Objectives:** Past proposal of similar objective. Previous objective was to finish DES SN Cosmology analysis and build infrastructure for LSST DESC SN Cosmology. We finished the DES SN Cosmology analysis.

***Statement of Potential Overlap:** No overlap concern as part of the same work.

***Proposal/Active Project Title:** Determining Systematics Uncertainties in Type Ia Supernova Analyses of DES and LSST with both Image and Catalog- Level Simulations

***Status of Support:** Current

Proposal/Award Number: DE-SC0021962

***Source of Support:** The Department of Energy

***Primary Place of Performance:** Duke University, Durham, North Carolina, United States

***Proposal/Active Project Start Date: (MM/YYYY):** 09/2021

***Proposal/Active Project End Date: (MM/YYYY):** 08/2026

***Total Anticipated Proposal/Project Amount:** \$750,000

*** Person Months per budget period Devoted to the Proposal/Active Project:**

Year	Person Months
2021	1
2022	1
2023	1
2024	1
2025	1

***Overall Objectives:** The overall objectives were focused on finishing up the DES SN cosmology analysis, and doing the first photo-z SN cosmology analysis (using DES). The latter years will be focused on applying these applications for LSST.

***Statement of Potential Overlap:** This new proposal is focused more on normal infrastructure than my DOE Career which was supposed to be innovative in what it figured out. As some of that innovation has been successful, it is becoming normal infrastructure. However, the work I hope to do as part of this proposal will be more true infrastructure, like pushing the difference imaging through, doing calibration, and getting from pixels-to-light curves. We have one new project with peculiar velocities, but more of it is focused on just getting a first Hubble diagram.

***Proposal/Active Project Title:** Big Data and the Biggest Tension in Cosmology

***Status of Support:** Current

Proposal/Award Number:

***Source of Support:** Packard Foundation

***Primary Place of Performance:** Duke University, Durham, North Carolina, United States

***Proposal/Active Project Start Date: (MM/YYYY):** 11/2019

***Proposal/Active Project End Date: (MM/YYYY):** 04/2025

***Total Anticipated Proposal/Project Amount:** \$875,000

*** Person Months per budget period Devoted to the Proposal/Active Project:**

Year	Person Months
2024	1

***Overall Objectives:** The main objective is to address systematic uncertainties in the local distance ladder. This project will end in the first year of the grant.

***Statement of Potential Overlap:** There may be very little time of overlap due to when the Packard ends. This proposal is more focused on local distance measures whereas the DOE work is more focused on DOE experiments (LSST and DES).

***Proposal/Active Project Title:** DEBASS Proposal

***Status of Support:** Pending

Proposal/Award Number:

***Source of Support:** NSF

***Primary Place of Performance:** Duke University, Durham, North Carolina, United States

***Proposal/Active Project Start Date: (MM/YYYY):** 06/2026

***Proposal/Active Project End Date: (MM/YYYY):** 06/2028

***Total Anticipated Proposal/Project Amount:** \$53,870

*** Person Months per budget period Devoted to the Proposal/Active Project:**

Year	Person Months
2026	0.5
2027	0.5
2028	0.5

***Overall Objectives:** Analyze the DEBASS sample and use it as a low-redshift anchor.

***Statement of Potential Overlap:** This is the only project working on the DEBASS sample.

***Proposal/Active Project Title:** Research in High Energy Physics at Duke University (Task C)

***Status of Support:** Pending

Proposal/Award Number:

***Source of Support:** DOE

***Primary Place of Performance:** Duke University, Durham, North Carolina, United States

***Proposal/Active Project Start Date: (MM/YYYY):** 04/2025

***Proposal/Active Project End Date: (MM/YYYY):** 04/2027

***Total Anticipated Proposal/Project Amount:** \$1,095,000

*** Person Months per budget period Devoted to the Proposal/Active Project:**

Year	Person Months
2025	1
2026	1

Year	Person Months
2027	1

***Overall Objectives:** This proposal: To do Y1 supernova cosmology with LSST DESC.

***Statement of Potential Overlap:** None with DEBASS.

Certification:

I certify that the information provided is current, accurate, and complete. This includes, but is not limited to, information related to current, pending, and other support (both foreign and domestic) as defined in 42 U.S.C. § 6605.

In accordance with Section 10632 of the CHIPS and Science Act of 2022 (42 U.S.C. § 19232), each individual identified as a senior/key person must certify that they are not a party to a malign foreign talent recruitment program.

Research Security Training Requirement for Federal Award Personnel: In accordance with Section 10634 of the CHIPS and Science Act of 2022 (42 U.S.C. § 19234), each individual identified as a senior/key person must certify that they have completed the requisite research security training that meets the requirements specified in Item 2 of Important Notice No. 149 within 12 months prior to proposal submission.

Certified by Scolnic, Daniel in SciENCv on 2026-01-17 15:37:25

CURRENT AND PENDING (OTHER) SUPPORT INFORMATION

Provide the following information for the Senior/key personnel and other significant contributors.
Follow this format for each person.

*NAME: Denton, Peter

PERSISTENT IDENTIFIER (PID): <https://orcid.org/0000-0002-5209-872X>

*POSITION TITLE: Physicist

*POSITION TITLE: Physicist

*ORGANIZATION AND LOCATION: Brookhaven National Lab, Upton, New York, United States

*ORGANIZATION AND LOCATION: Brookhaven National Lab, Upton, New York, United States

Proposals and Active Projects

***Proposal/Active Project Title:** Brookhaven Lab

***Status of Support:** Current

Proposal/Award Number: DE-SC0012704

***Source of Support:** Department of Energy

***Primary Place of Performance:** Brookhaven National Lab

***Proposal/Active Project Start Date: (MM/YYYY):** 10/2025

***Proposal/Active Project End Date: (MM/YYYY):** 09/2026

***Total Anticipated Proposal/Project Amount:** \$884,000,000

*** Person Months per budget period Devoted to the Proposal/Active Project:**

Year	Person Months
2025	2
2026	10

***Overall Objectives:** The budget funds the entire lab continuously. In my group, this is to carry out high energy theory research.

***Statement of Potential Overlap:** There is no overlap with the proposal to study supernova followup techniques with Vera Rubin as this, for me, focuses on neutrino oscillation theory.

***Proposal/Active Project Title:** Implementing Rubin Galactic Supernova Trigger from Super-Kamiokande

***Status of Support:** Pending

Proposal/Award Number:

***Source of Support:** US Department of Energy

***Primary Place of Performance:** Brookhaven National Lab

***Proposal/Active Project Start Date: (MM/YYYY):** 10/2026

***Proposal/Active Project End Date: (MM/YYYY):** 09/2028

***Total Anticipated Proposal/Project Amount:** \$200,000

*** Person Months per budget period Devoted to the Proposal/Active Project:**

Year	Person Months
2026	0.1
2027	0.1

***Overall Objectives:** Develop a pipeline to connect a neutrino alert of a galactic supernova to Vera Rubin Observatory for rapid study.

***Statement of Potential Overlap:** This does not overlap with my base funding as my base funding is to do high energy particle physics research and this is to do astrophysics.

Addendum

Do the Notice of Funding Opportunity (NOFO) or Award instructions require a Current and Pending (Other) Support addendum?

Yes

1. Have you started or spun-out any companies, whether foreign, domestic, publicly traded, or non-publicly traded?

No

2. Do you have an affiliation with an institution located in an FCOC?

No

3. Within the past seven years, have you collaborated with individuals at an FCOC site or used facilities or instrumentation at an FCOC site? Collaborations do not include routine workshops or conferences hosted in FCOCs.

I have written a theory paper on neutrino physics with a graduate student in China. No resources were shared in this project.

4. Do you currently serve, or have you ever served, in the military or intelligence service or

police force of an FCOC?

No

5. Within the past seven years, have you been offered any of the following by an FCOC: a. Future compensation; b. Complimentary foreign travel, lodging, and/or per diem; c. Things of non de minimus value (i.e., when aggregated, exceeds \$5,000); d. Honorific titles; e. Career advancement opportunities; f. Other types of remuneration or consideration

No

6. Within the past seven years have you received any FCOC sources of support, whether from a public or private entity or individual? Sources of support include:

- a. Salary and any payment for services not otherwise identified as salary (e.g., consulting fees, honoraria, paid authorship);**
- b. Equity interest (e.g., any stock, stock option, or other ownership interest);**
- c. Income related to intellectual property rights and interests (e.g., patents, copyrights);**
- d. Reimbursed or sponsored travel (i.e., that which is paid on behalf of the covered individual and not reimbursed to the covered individual so that the exact monetary value may not be readily available);**
- e. Current and pending (other) support as defined in the National Security Presidential Memorandum-33 Implementation Guidance Appendix Definitions;**
- f. Non-monetary support of goods, services, or expertise;**
- g. Payments from an FCOC to family or friends on the covered individual's behalf or held in any other accounts on their behalf (promise of future access).**

No

7. To the best of your knowledge, are you an inventor on a patent application filed in an FCOC, without a companion US patent application?

No

Certification:

I understand that I have been designated as a covered individual by the Federal funding agency.

I certify to the best of my knowledge and belief that the information contained in this Current and Pending (Other) Support Form is true, complete, and accurate. I understand that any false, fictitious, or fraudulent information, misrepresentations, half-truths, or omissions of any material fact, may subject me to criminal, civil or administrative penalties for fraud, false statements, false claims or otherwise. (18 U.S.C. §§ 1001 and 287, and 31 U.S.C. 3729-3733 and 3801-3812). I further understand and agree that (1) the statements and representations made herein are material to DOE's funding decision, and (2) I have a responsibility to update the disclosures during the project period of the award should circumstances change which impact the responses provided above.

I also certify that, at the time of submission, I am not a party in a malign foreign talent recruitment program. I further understand should I take action to involve myself with a Malign Foreign Talent Recruitment Program during the period of performance of the award, I must notify the recipient's Authorized Agent immediately, but no later than five business days of taking such action and immediately recuse myself from all DOE awards.

If this is an R&D proposal/award, I also certify to the following statement:

Within the past 12 months I have completed research security training meeting the requirements in SEC. 10634(b) of 42 USC 19234.

Certified by Denton, Peter in SciENCv on 2026-01-21 15:07:07

CURRENT AND PENDING (OTHER) SUPPORT INFORMATION

Provide the following information for the Senior/key personnel and other significant contributors.
Follow this format for each person.

*NAME: Blum, Robert David

PERSISTENT IDENTIFIER (PID): <https://orcid.org/0000-0002-8622-4237>

*POSITION TITLE: Rubin Observatory Director for Operations

*ORGANIZATION AND LOCATION: AURA/NSFs NOIRLab, Tucson, Arizona, United States

Proposals and Active Projects

***Proposal/Active Project Title:** NSF's National Optical-Infrared Astronomy Research Laboratory (NOIRLab): Management and Operations of Vera C. Rubin Observatory

***Status of Support:** Current

Proposal/Award Number: 2211468

***Source of Support:** NSF/CSA

***Primary Place of Performance:** Tucson, AZ

***Proposal/Active Project Start Date: (MM/YYYY):** 10/2023

***Proposal/Active Project End Date: (MM/YYYY):** 09/2027

***Total Anticipated Proposal/Project Amount:** \$163,601,802

* **Person Months per budget period Devoted to the Proposal/Active Project:**

Year	Person Months
2023	12
2024	12
2025	12
2026	12
2027	12

***Overall Objectives:** i. In pre-operations: (a) Team ramp-up to sufficient staffing levels and competencies to enable a smooth transition from construction to full operations and ensure seamless knowledge transfer, including training in key procedures on the fully integrated system. (b) Coordinate closely with the construction and commissioning teams to transition staff into operations. (c) Develop and implement key operational processes, functions, and communication channels to ensure operational preparedness. (d) Respond to needed revisions in science algorithms, etc based on

lessons learned from commissioning data. ii. In survey operations: (a) Maintain and operate the Rubin Summit Facility and coordinate closely with SLAC in the maintenance and operation of the LSST Camera. (b) Maintain and operate the Rubin Base Facility including the Chilean Data Center. (c) Coordinate closely with SLAC in the portions of the maintenance and operation of the U.S. Data Facility. (d) Maintain and operate the long-haul networks and coordinate closely with SLAC on the maintenance and operation of the long-haul network portions funded by DOE. (e) Maintain and operate the remaining portions of the Rubin Observatory System funded by NSF. (f) Maintain operational processes, functions, and communication channels to assure successful execution operations. (g) Respond to needed revisions in science algorithms, pipelines etc. (h) Manage the in-kind contribution program and data rights holders.

***Statement of Potential Overlap:** N/A

***Proposal/Active Project Title:** NSF's National Optical-infrared Astronomy Research Laboratory: Pre-Operations of the Vera C. Rubin Observatory

***Status of Support:** Current

Proposal/Award Number: 1836783

***Source of Support:** NSF/CSA

***Primary Place of Performance:** Tucson, AZ

***Proposal/Active Project Start Date: (MM/YYYY):** 10/2019

***Proposal/Active Project End Date: (MM/YYYY):** 09/2026

***Total Anticipated Proposal/Project Amount:** \$17,188,735

*** Person Months per budget period Devoted to the Proposal/Active Project:**

Year	Person Months
2019	12
2020	12
2021	12
2022	12

***Overall Objectives:** i. Team ramp-up to sufficient staffing levels and competencies to enable a smooth transition from construction to full operations and ensure seamless knowledge transfer, including training in key procedures on the fully integrated system. ii. Coordinate closely with the construction and commissioning teams to transition staff into operations for a smooth team ramp-up. iii. Develop and implement key operational processes, functions, and communication channels to ensure operational preparedness for Rubin Observatory's high flux of data. iv. Respond to needed revisions in science algorithms, data pipeline, and data distribution based on lessons learned from commissioning data. 2023-2026 are supported by a no cost extension.

***Statement of Potential Overlap: N/A**

Addendum

Do the Notice of Funding Opportunity (NOFO) or Award instructions require a Current and Pending (Other) Support addendum?

No

Certification:

I understand that I have been designated as a covered individual by the Federal funding agency.

I certify to the best of my knowledge and belief that the information contained in this Current and Pending (Other) Support Form is true, complete, and accurate. I understand that any false, fictitious, or fraudulent information, misrepresentations, half-truths, or omissions of any material fact, may subject me to criminal, civil or administrative penalties for fraud, false statements, false claims or otherwise. (18 U.S.C. §§ 1001 and 287, and 31 U.S.C. 3729-3733 and 3801-3812). I further understand and agree that (1) the statements and representations made herein are material to DOE's funding decision, and (2) I have a responsibility to update the disclosures during the project period of the award should circumstances change which impact the responses provided above.

I also certify that, at the time of submission, I am not a party in a malign foreign talent recruitment program. I further understand should I take action to involve myself with a Malign Foreign Talent Recruitment Program during the period of performance of the award, I must notify the recipient's Authorized Agent immediately, but no later than five business days of taking such action and immediately recuse myself from all DOE awards.

If this is an R&D proposal/award, I also certify to the following statement:

Within the past 12 months I have completed research security training meeting the requirements in SEC. 10634(b) of 42 USC 19234.

Certified by Blum, Robert in SciENCv on 2026-01-17 10:44:47

CURRENT AND PENDING (OTHER) SUPPORT INFORMATION

Provide the following information for the Senior/key personnel and other significant contributors.
Follow this format for each person.

*NAME: Thomas, Sandrine

PERSISTENT IDENTIFIER (PID): <https://orcid.org/0000-0002-9121-3436>

*POSITION TITLE: Deputy Director / Telescope and Site Scientist

*ORGANIZATION AND LOCATION: AURA / Vera C. Rubin Observatory, Tucson, Arizona, United States

Proposals and Active Projects

***Proposal/Active Project Title:** NSF's National Optical-Infrared Astronomy Research Laboratory (NOIRLab): Management and Operations of Vera C. Rubin Observatory

***Status of Support:** Current

Proposal/Award Number: 2211468

***Source of Support:** NSF/CSA

***Primary Place of Performance:** Tucson

***Proposal/Active Project Start Date: (MM/YYYY):** 10/2023

***Proposal/Active Project End Date: (MM/YYYY):** 09/2027

***Total Anticipated Proposal/Project Amount:** \$163,601,802

* **Person Months per budget period Devoted to the Proposal/Active Project:**

Year	Person Months
2026	12
2027	12

***Overall Objectives:** i. In pre-operations: (a) Team ramp-up to sufficient staffing levels and competencies to enable a smooth transition from construction to full operations and ensure seamless knowledge transfer, including training in key procedures on the fully integrated system. (b) Coordinate closely with the construction and commissioning teams to transition staff into operations. (c) Develop and implement key operational processes, functions, and communication channels to ensure operational preparedness. (d) Respond to needed revisions in science algorithms, etc based on lessons learned from commissioning data. ii. In survey operations: (a) Maintain and operate the Rubin Summit Facility and coordinate closely with SLAC in the maintenance and operation of the LSST Camera. (b) Maintain and operate the Rubin Base Facility including the Chilean Data Center. (c) Coordinate closely with SLAC in the portions of the maintenance and operation of the U.S. Data

Facility. (d) Maintain and operate the long-haul networks and coordinate closely with SLAC on the maintenance and operation of the long-haul network portions funded by DOE. (e) Maintain and operate the remaining portions of the Rubin Observatory System funded by NSF. (f) Maintain operational processes, functions, and communication channels to assure successful execution operations. (g) Respond to needed revisions in science algorithms, pipelines etc. (h) Manage the inkind contribution program and data rights holders.

***Statement of Potential Overlap: NA**

Addendum

Do the Notice of Funding Opportunity (NOFO) or Award instructions require a Current and Pending (Other) Support addendum?

No

Certification:

I understand that I have been designated as a covered individual by the Federal funding agency.

I certify to the best of my knowledge and belief that the information contained in this Current and Pending (Other) Support Form is true, complete, and accurate. I understand that any false, fictitious, or fraudulent information, misrepresentations, half-truths, or omissions of any material fact, may subject me to criminal, civil or administrative penalties for fraud, false statements, false claims or otherwise. (18 U.S.C. §§ 1001 and 287, and 31 U.S.C. 3729-3733 and 3801-3812). I further understand and agree that (1) the statements and representations made herein are material to DOE's funding decision, and (2) I have a responsibility to update the disclosures during the project period of the award should circumstances change which impact the responses provided above.

I also certify that, at the time of submission, I am not a party in a malign foreign talent recruitment program. I further understand should I take action to involve myself with a Malign Foreign Talent Recruitment Program during the period of performance of the award, I must notify the recipient's Authorized Agent immediately, but no later than five business days of taking such action and immediately recuse myself from all DOE awards.

If this is an R&D proposal/award, I also certify to the following statement:

Within the past 12 months I have completed research security training meeting the requirements in SEC. 10634(b) of 42 USC 19234.

Certified by Thomas, Sandrine in SciENCv on 2026-01-22 07:54:26

CURRENT AND PENDING (OTHER) SUPPORT INFORMATION

Provide the following information for the Senior/key personnel and other significant contributors.
Follow this format for each person.

*NAME: Roodman, Aaron

PERSISTENT IDENTIFIER (PID): <https://orcid.org/0000-0001-5326-3486>

*POSITION TITLE: Professor of Particle Physics & Astrophysics

*ORGANIZATION AND LOCATION: SLAC National Accelerator Laboratory, Menlo Park, California, USA

Proposals and Active Projects

*Proposal/Active Project Title: AI/ML Cosmic Frontier Research

*Status of Support: Current

Proposal/Award Number:

*Source of Support: DOE Office of HEP

*Primary Place of Performance: SLAC National Accelerator Laboratory

*Proposal/Active Project Start Date: (MM/YYYY): 10/2025

*Proposal/Active Project End Date: (MM/YYYY): 10/2026

*Total Anticipated Proposal/Project Amount: \$200,000

* Person Months per budget period Devoted to the Proposal/Active Project:

Year	Person Months
2025	4
2026	4

*Overall Objectives: This is funding for Cosmic Frontier AI/ML, and the research scope here is cosmology with strong gravitational lenses, making use of ML methods such as Simulation-Based Inference.

*Statement of Potential Overlap: This research makes use of ML methods for Rubin Observatory. It is focussed on strong lenses found and studied with Rubin Observatory. As such there is no direct overlap with this proposal.

***Proposal/Active Project Title:** FPD Directorate

***Status of Support:** Current

Proposal/Award Number:

***Source of Support:** DOE Office of HEP

***Primary Place of Performance:** SLAC National Accelerator Laboratory

***Proposal/Active Project Start Date: (MM/YYYY):** 10/2025

***Proposal/Active Project End Date: (MM/YYYY):** 09/2026

***Total Anticipated Proposal/Project Amount:** \$150,000

*** Person Months per budget period Devoted to the Proposal/Active Project:**

Year	Person Months
2025	3
2026	3

***Overall Objectives:** Support in my role as Deputy Associate Laboratory Director for the Fundamental Physics Directorate

***Statement of Potential Overlap:** None

***Proposal/Active Project Title:** Rubin Observatory Research in support of Operations

***Status of Support:** Current

Proposal/Award Number:

***Source of Support:** DOE Office of HEP

***Primary Place of Performance:** SLAC National Accelerator Laboratory

***Proposal/Active Project Start Date: (MM/YYYY):** 10/2025

***Proposal/Active Project End Date: (MM/YYYY):** 10/2026

***Total Anticipated Proposal/Project Amount:** \$200,000

*** Person Months per budget period Devoted to the Proposal/Active Project:**

Year	Person Months
2025	4
2026	4

***Overall Objectives:** Operation of the LSST Camera and the Vera Rubin Observatory. This funding

is part of SLAC's Cosmic Frontier research budget.

***Statement of Potential Overlap:** There is significant synergy between my work supporting the operation of the LSST Camera and Rubin Observatory and this proposal. This proposal represents new scope not currently undertaken with my research in support of operations.

***Proposal/Active Project Title:** HEP Traineeship at SJSU

***Status of Support:** Current

Proposal/Award Number:

***Source of Support:** Office of Science, HEP

***Primary Place of Performance:** SLAC

***Proposal/Active Project Start Date: (MM/YYYY):** 10/2024

***Proposal/Active Project End Date: (MM/YYYY):** 09/2026

***Total Anticipated Proposal/Project Amount:** \$7,500

*** Person Months per budget period Devoted to the Proposal/Active Project:**

Year	Person Months
2025	0.12
2026	0.12

***Overall Objectives:** Support HEP RENEW program at SJSU with coordination and recruitment of mentors at SLAC.

***Statement of Potential Overlap:** None.

***Proposal/Active Project Title:** LDRD Prototype fiber-feed for a Rubin Multi-Object Spectrograph

***Status of Support:** Current

Proposal/Award Number:

***Source of Support:** SLAC National Accelerator Laboratory

***Primary Place of Performance:** SLAC

***Proposal/Active Project Start Date: (MM/YYYY):** 09/2024

***Proposal/Active Project End Date: (MM/YYYY):** 10/2026

***Total Anticipated Proposal/Project Amount:** \$15,000

*** Person Months per budget period Devoted to the Proposal/Active Project:**

Year	Person Months
2025	0.5
2026	0.5

***Overall Objectives:** The purpose of this LDRD is to develop or demonstrate workable optical designs for a multi-object fiber system on the fast F/1.2 Rubin Observatory optical beam.

***Statement of Potential Overlap:** No overlap with this proposal

Addendum

Do the Notice of Funding Opportunity (NOFO) or Award instructions require a Current and Pending (Other) Support addendum?

No

Certification:

I understand that I have been designated as a covered individual by the Federal funding agency.

I certify to the best of my knowledge and belief that the information contained in this Current and Pending (Other) Support Form is true, complete, and accurate. I understand that any false, fictitious, or fraudulent information, misrepresentations, half-truths, or omissions of any material fact, may subject me to criminal, civil or administrative penalties for fraud, false statements, false claims or otherwise. (18 U.S.C. §§ 1001 and 287, and 31 U.S.C. 3729-3733 and 3801-3812). I further understand and agree that (1) the statements and representations made herein are material to DOE's funding decision, and (2) I have a responsibility to update the disclosures during the project period of the award should circumstances change which impact the responses provided above.

I also certify that, at the time of submission, I am not a party in a malign foreign talent recruitment program. I further understand should I take action to involve myself with a Malign Foreign Talent Recruitment Program during the period of performance of the award, I must notify the recipient's Authorized Agent immediately, but no later than five business days of taking such action and immediately recuse myself from all DOE awards.

If this is an R&D proposal/award, I also certify to the following statement:

Within the past 12 months I have completed research security training meeting the requirements in SEC. 10634(b) of 42 USC 19234.

Certified by Roodman, Aaron in SciENCv on 2026-01-26 12:06:16

Appendix 4: Bibliography and References Cited

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Appendix 7: Data Management Plan

This project will not produce any data. It's focus is on developing the neutrino trigger that will enable Target-of-Opportunity observation by the Rubin Observatory based on trigger from Super-Kamiokande. Data from Rubin and Super-Kamiokande will remain subject to their respective data policies. For completeness, we list the LSST and Super-Kamiokande data management policies below.

Rubin Observatory DMP: <https://docushare.lsstcorp.org/docushare/dsweb/Get/LPM-151>

LSST DESC DMP: https://lsstdesc.org/assets/pdf/docs/DESC_DMP_latest.pdf

Super-Kamiokande DMP:

Data Types and Sources

SK detector raw data is comprised of binary information that is readable by SK proprietary code which relies on data bases to properly identify active channels, apply calibration constants, and further organize the information for analysis. Event data is largely background from cosmic rays and radioactive decays, and must be processed by reduction software to select events of interest. Events of interest are then processed further by detailed reconstruction algorithms that identify the characteristics of the events. Equally as important as detector data are large Monte Carlo simulations of various processes that may be observed in the detector data. Likewise, dedicated calibration data are used to understand the performance of the detector.

Reconstructed events are used to perform physics analyses such as neutrino observations and the search for proton decay. The results of these analyses are published in peer-reviewed journals.

Content and Format

The format of the event raw data as well as the reduced and reconstructed data are binary files that require proprietary code the fully access and interpret. Subsequent analysis uses summary information stored in the CERN ROOT or ntuple format. The data analysis code is preserved using a versioning and archiving system, currently SVN.

Preservation and Sharing

The SK experiment has undergone several phases (SK-I, II, III, IV, V, and SK-Gd), and has collected data since 1996. The SK collaboration treats the collected data carefully and has preserved the raw data on tape backup since the beginning of the SK experiment. The SK collaboration routinely performs new analyses and searches through the entire 20 year data set, generally starting with reduced and reconstructed files that are kept on redundant disk systems.

Due to the complexity of event level data, general access is not practical. Upon request, data tables, content of figures, chi-squared maps, and other information that facilitates sharing with non-SK researchers has been made available, and will continue to be made available.

Protection

SK data do not contain personal or confidential information and do not require special security measures.

Rationale

This data management plan is in line with current practices of large particle physics experiments.